

Predicting All-Cause Hospital Readmissions with NLP and Cox Modeling

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Business Problem

Although it may seem counterintuitive, hospital readmissions ultimately cost hospitals and the healthcare system as a whole. This is because hospitals do not always receive the full amount billed to patients. Reimbursements are usually negotiated with insurers, and payments are based on adjusted rates rather than an itemized bill. When a readmission is preventable, it creates additional costs through the repeated use of beds, staff time, tests, procedures, and care coordination. It can also reduce hospital revenue through quality-based payment adjustments, including the federal Hospital Readmissions Reduction Program, which “punishes general acute-care hospitals when more Medicare patients return for a new admission within 30 days of discharge than the government decides is appropriate” (Rau, 2021). This project uses NLP on an EHR dataset to support a Cox time-to-event model for all-cause hospital readmission prediction.

Background/History

This issue is becoming more pressing as the U.S. population ages, healthcare costs continue to rise, and increased clinician shortages place additional strain on hospitals. Older patients are more likely to have chronic conditions and to be readmitted. This phenomenon increases the operational and financial burden on health systems. “Recurrent re-hospitalizations are associated with progressive worsening of functional state, increased risk of institutionalization, and increased mortality” (Bortolani et al., 2024). Predicting readmission risk can help hospitals target limited resources toward patients most likely to return, reduce avoidable costs, and improve quality performance and patient outcomes.

Data Explanation

The dataset used for this project is the *Medical records of 30K Synthea synthetic patients* dataset from the Harvard Dataverse. This dataset was chosen because the project required a large dataset that realistically simulated patients’ electronic health records. The project required rich text for NLP, dates from which to calculate time to the event (in this case, hospital readmission within 30 days), and covariate

features for the Cox model, such as demographic or financial features. The dataset used is available for download via two separate zip files, each containing simulated data for about 15K patients.

Each zip file contains 13 individual CSV files with the following titles: allergies, careplans, conditions, devices, encounters, immunizations, medications, observations, organizations, patients, payers, procedures, and providers. I ended up only using four of the files: patients, conditions, encounters, and procedures, to create a master data frame that included information from both populations, with each row representing a single patient (Chen, 2022).

Methods

The master data frame was filtered to include only inpatient encounters, which created a population of 103,223 true hospital admissions. Missingness was addressed with meaningful imputations or omissions, depending on the feature and its relevance to the project. EDA revealed features with high cardinality and multicollinearity, properties that had to be resolved to satisfy the Cox Model's requirements. Features were removed to prevent overfitting and noise within the model.

Care was taken to aggregate the written description notes into a running history without introducing data leakage. The pandas library was used to scan backward in time to dynamically capture a patient's complete history of pre-existing conditions up to their admission day, helping aggregate meaningful information into the description features used to build the model. Feature selection was conducted on the aggregated description text using SentenceTransformer, a miniLM distilled from BERT.

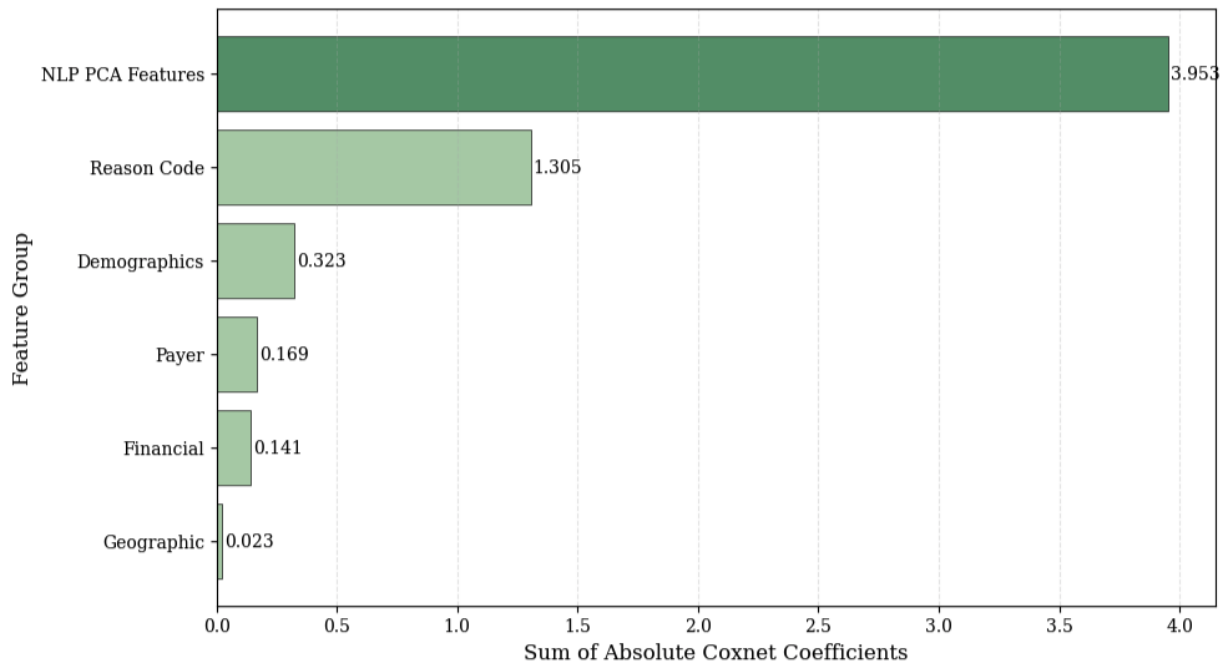
After creating the embeddings, PCA was used to compress the high-dimensional results to their most important features. It is important to note that PCA was applied to the full embedding matrix before train/test splitting due to computational constraints. Because PCA is unsupervised and does not use the readmission outcome, the risk of target leakage is limited. However, future work should fit PCA only on the training set and apply the learned transformation to the test set.

Once the PCA features were ready, the duration, event, and covariate features were defined in preparation for the Cox Model. The final survival dataset was collapsed to one row per patient. The first inpatient encounter was treated as the baseline encounter, and the event was defined as a readmission within 30 days of discharge from that encounter. The final master dataframe contained 11,792 patients, with an event rate of 8.5%. This means that 8.5% of the population was readmitted within 30 days of getting discharged (See Appendix A). The dataframe was then split into training and test sets, and all categorical features were one-hot encoded, and all numerical features were scaled.

The final model used was scikit-survival's CoxnetSurvivalAnalysis. A penalized model was selected because the unpenalized Cox model produced a singular matrix, likely due to high-dimensional features, correlated predictors, and one-hot encoded categorical variables.

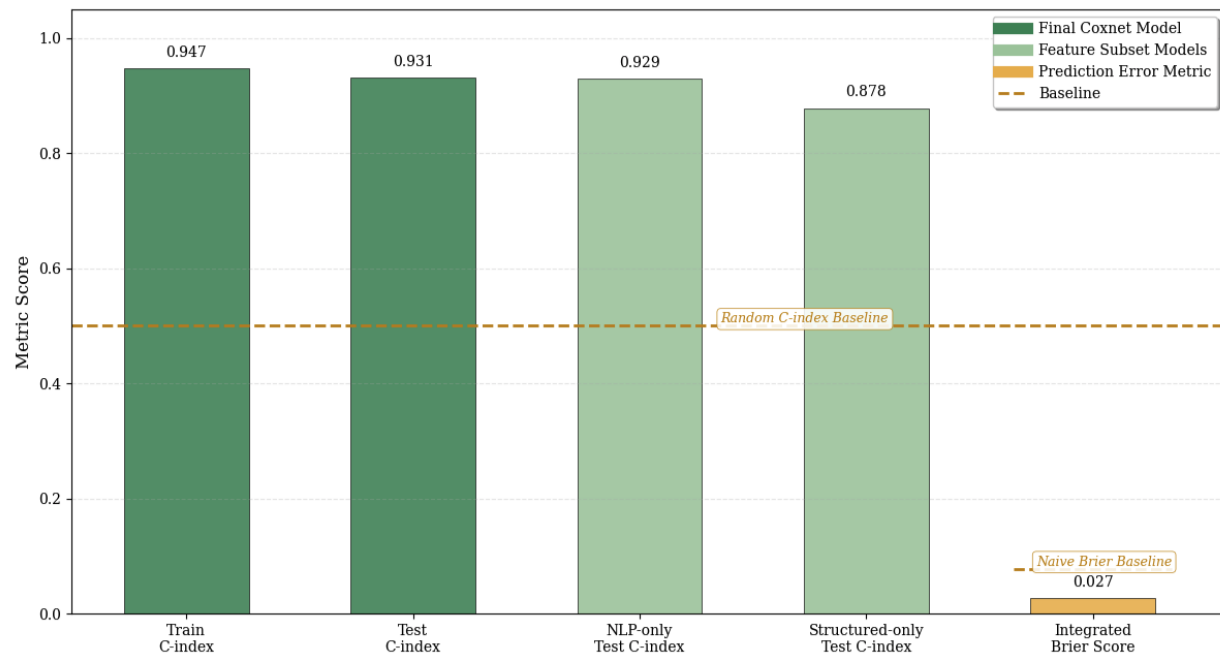
Analysis

Concordance Index (C-index), Brier Score, and Cumulative/dynamic AUC (as an exploratory metric) were used to evaluate the models. The final model had scores indicating strong discrimination on unseen patients (C-index train: 0.947, C-index test: 0.931). Because the scores were so high, feature-set comparison models were created to test for data leakage and overfitting. An NLP-only model achieved a similar test C-index of 0.929, while the structured-only model achieved a lower test C-index of 0.878. This suggests that the NLP-derived PCA features captured most of the predictive signal associated with 30-day readmission risk (see Figure 1).

Figure 1*Grouped Feature Contribution in Cox Readmissions Model*

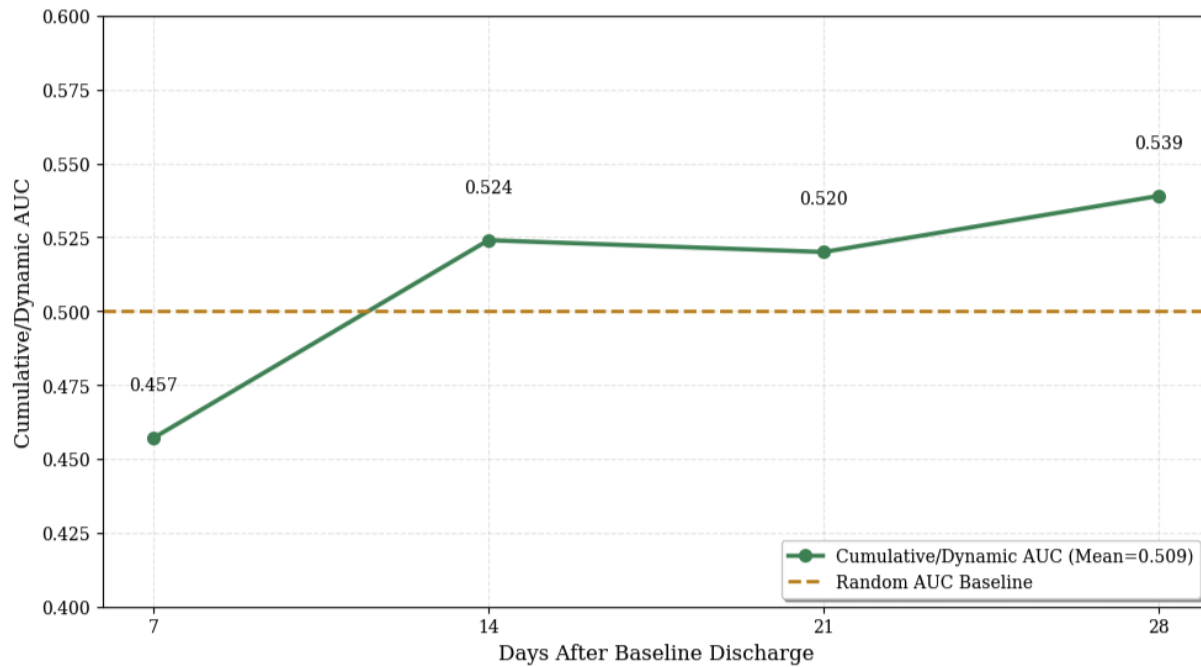
Note. Feature contribution was estimated using the summed absolute Coxnet coefficients by feature group. Because coefficients are model-based rather than causal, these values should be interpreted as relative predictive contributions rather than causal importance.

The penalized Cox model achieved a training C-index of 0.947 and a test C-index of 0.931, indicating strong discrimination in unseen patients. The Integrated Brier Score was 0.027 across days 7, 14, 21, and 28, suggesting low prediction error over the 30-day readmission window (see Figure 2).

Figure 2*Cox Hospital Readmissions Model Evaluation Metrics*

Note. The final model's train and test C-index, as well as the feature set comparison models, were well above the 0.5 random C-index baseline threshold. The Integrated Brier Score for the final model was also below the Naive Brier Baseline, indicating that the models had low predictive error and ranked patient readmission times strongly.

The cumulative/dynamic AUC was unstable because the study design used 30-day censoring. Most non-readmitted patients were censored at the same point in time (at exactly 30 days), leading to a poorly estimated censoring distribution for inverse probability weighting. As a result, cumulative/dynamic AUC was treated as exploratory rather than a primary evaluation metric (see Figure 3).

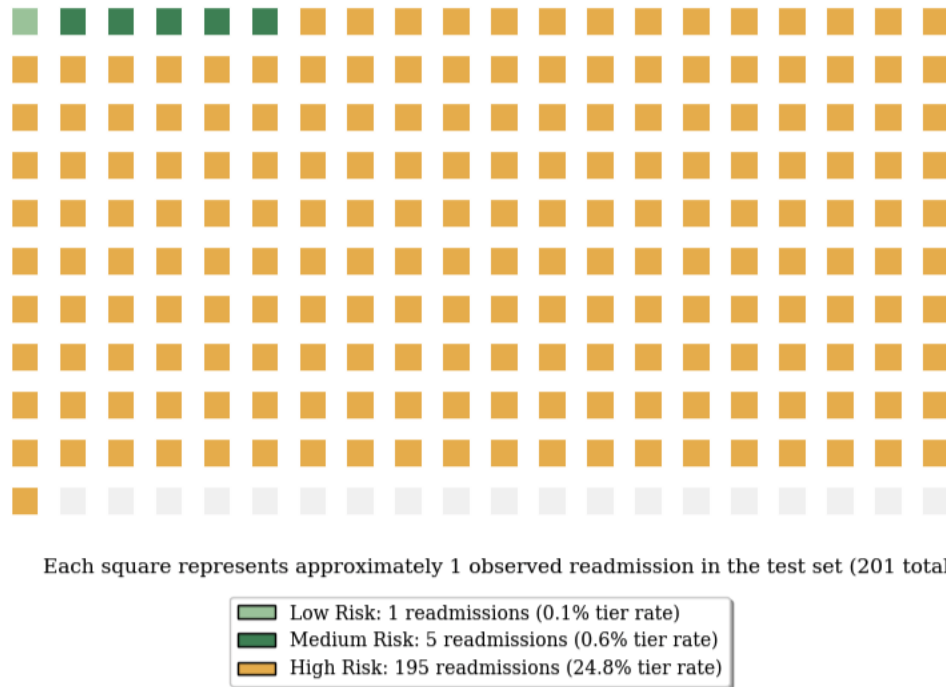
Figure 3*Cox Hospital Readmissions Cumulative/Dynamic AUC*

Note. The Cumulative/Dynamic AUC values hover around the random AUC baseline due to instability caused by the 30-day administrative censoring.

Predicted Coxnet risk scores were used to create low-, medium-, and high-risk tiers. Patients were divided into tertiles based on predicted risk scores. The low-risk tier had 786 patients and 1 readmission, for an event rate of 0.1%. The medium-risk tier had 786 patients and 5 readmissions, for an event rate of 0.6%. The high-risk tier had 787 patients and 195 readmissions, for an event rate of 24.8%. The high-risk tier captured approximately 97% of observed test-set readmissions (see Figure 4).

Figure 4

Where 30-Day Readmissions Occurred Across Risk Tiers



Note. The tier distribution demonstrates the value of correctly identifying the patients with the highest readmission risk.

Conclusion

This project demonstrates that NLP-enhanced EHR features can meaningfully support hospital readmission prediction using penalized Cox modeling. The final Coxnet model achieved strong discrimination and low prediction error, while the NLP-only model nearly matched the full model's performance. This suggests that narrative-style clinical information contained a substantial predictive signal.

The final risk-tier analysis supports the model's practical value. Patients in the high-risk tier had a much higher observed readmission rate than patients in the low- and medium-risk tiers, suggesting that

the model could help identify patients who may benefit from additional discharge planning or follow-up support.

Assumptions

This project assumes that the synthetic EHR data is realistic enough to support a proof-of-concept readmission model. It also assumes that the first inpatient encounter can reasonably serve as a baseline encounter and that a 30-day readmission window is appropriate for the business problem.

Because the model uses a Cox-based framework, it also relies on the general logic of proportional hazards modeling. However, because the final model was penalized and high-dimensional, the project focused more on predictive performance than direct clinical interpretation of individual coefficients.

Limitations/Challenges

The largest limitation is that the dataset is synthetic. Synthetic data protects privacy and allows modeling work without direct exposure to real patient records, but the results are only as reliable as the simulation process. Relationships in synthetic data may be cleaner, stronger, or less complex than relationships in real-world EHR data. Because of that, more research would need to be done with real patient data through the same pipeline.

There were many challenges with this project. One of the stand-out challenges was how long the NLP step took. Even though I tried to reduce the computational load by deduplicating common phrases, the step still took about twenty minutes on my personal computer. Using a platform with more compute would have sped up the process.

Future Use/Additional Applications

Future use of the same pipeline could be applied to real-world EHR data, such as physicians' notes, patient discharge summaries, medication histories, or lab results. The pipeline could be modified to predict a range of events in the healthcare space, such as time to death, time to ER return, or time to ICU transfer.

Recommendations

For hospitals interested in reducing readmissions to both save money and improve patient outcomes, they should consider using structured EHR data and clinical text. This project suggests that text features can contain strong predictive power. However, since the project is conducted with synthetic data, results would need to be validated on a real-world dataset before being relied upon.

Once the pipeline has been validated using real-world data, risk scores can be used to prioritize patient care resources, schedule follow-up calls, conduct discharge planning, or to schedule earlier outpatient appointments.

Implementation Plan

Once the model has been validated on real-world data, the pipeline can be implemented using big data framework, such as NiFi, Apache Kafka, Apache Spark, HDFS, HBase, Apache Hive, Apache Solr, and a dashboard or direct to hospital's charting software (such as Epic or Cerner). The output would be most effective if it's incorporated in existing workflows. Periodic retraining and monitoring for model drift will be required as care techniques and disease responses change over time.

Ethical Assessment

This project uses synthetic data, which reduces privacy risk because it does not directly expose real patient records. However, synthetic data can still reflect biases present in the real-world data or assumptions used to generate it. If biased patterns exist in the simulation process, they may carry over into the model. It has been recommended that the pipeline be used on real-world EHR data, but the model will only succeed if each demographic is carefully considered.

References

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<https://huggingface.co/sentence-transformers>

Appendix A

Duration and Event Variables

```

                                patient      start \
0  0005b429-ed45-a435-fc39-4f395cf5833a 1993-05-29 19:12:19
1  0007efd3-c50d-7e51-6a30-8dc08d4269fc 2021-01-05 08:05:10
2  000eba84-0d5c-2657-0b5e-0c70f2f746c2 2021-04-19 01:41:09
3  002ba0b7-6708-9c6f-741c-0c3934957366 2016-09-02 09:08:47
4  002bc307-2fff-04ba-161b-98cce123e226 2012-01-19 19:30:33

                                stop  time_to_readmit  event      age
0  1993-05-30 19:12:19                30.0          0  64.386037
1  2021-01-15 08:24:20                30.0          0  57.114305
2  2021-04-20 01:41:09                30.0          0  92.035592
3  2016-09-03 09:08:47                30.0          0  31.797399
4  2012-01-20 19:30:33                30.0          0  78.433949

Final patient-level survival shape: (11792, 84)
Unique patients: 11792
Event rate: 0.085

```

Note. In a Cox Model, duration refers to the time until the event or until censoring, represented as `time_to_readmit` in this table. The event, in this case, is hospital readmission. An event flag was created, with 0 representing no readmission and 1 representing readmission.